MCO-1D ORNITHOPTER

Bill of Materials — Revision 15 — July 2026

This BOM reflects the complete corrected architecture: 8 independent motor-sector assemblies (one GoolRC 5010 motor per wing, no shared drives), 4 × ODrive v3.6 boards (axis0=port, axis1=starboard per pair), 2 × MCP3208 ADC chips for per-wing pitch and sweep feedback, and the full 45,229-line Wave Library controlling everything at 50 Hz.

Aircraft	MCO-1D metachronal cascade ornithopter
MTOW	3.448 kg (7.60 lb)
Wings	8 — four tandem pairs W1 (front) to W4 (rear)
Motors	8 × GoolRC 5010-14 360KV — one per wing, independent
ODrive boards	4 × v3.6 — axis0 = port wing, axis1 = starboard
Pitch servos	8 × TORQ TCM-1001 — AoA control, feedback pin required
Sweep servos	8 × DS3225 — fore-aft position, feedback pin required
Pitch feedback ADC	MCP3208 on SPI1 CE1 — actual AoA per wing at 50 Hz
Sweep feedback ADC	MCP3208 on SPI1 CE0 — actual sweep per wing
Stroke sensing	ODrive pos_estimate CAN — no new hardware
Flight computer	Raspberry Pi 4 — 4 GB RAM
Wave Library	45,229 lines — 29 hypotheses, 109 variants
Document version	Revision 15 — July 2026

1. AIRFRAME AND STRUCTURE

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	Main airframe / fuselage	Composite or carbon fibre. Must mount 8 sector wheel axles, 8 motor brackets, 16 servo brackets, 4 ODrive boards, Pi, all electronics. Vibration-resistant mounting for all PCBs.	Custom
8	pcs	Wing panels — S1223 aerofoil, chord 165 mm	Lightweight composite or foam-film. Each wing independent. Must withstand 96° amplitude at 3.0 Hz repeated cycling. Span: W1=772 mm, W2=749 mm, W3=735 mm, W4=712 mm (semi-span).	Custom
8	pcs	Sector wheel assemblies — 100° mechanical arc	One per wing. Converts motor rotation to flapping arc. Hard stop at 100°. Software limits to 96° (4° safety margin). Must rotate smoothly across full arc under aerodynamic load.	Custom
8	sets	Rope drive — one per wing, independent	IMPORTANT: each wing has its own rope drive. There is NO shared rope between port and starboard on any pair. Motor drum → rope → sector wheel drum. Inspect before every flight.	Cord
8	pcs	Wing pivot and AoA linkage hardware	Pitch servo horn connects to wing leading edge. Must allow ±22° AoA rotation without binding.	Hardware
1	set	Landing skids or float mount points	Land ops: skids. Water ops (future): float attachment hardware TBD after TC-02 hover confirmation.	Custom

2. MOTORS — 8 INDEPENDENT ASSEMBLIES

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	GoolRC 5010-14 Brushless Motor — 360 KV	12N14P configuration. 7 pole pairs (verified). ONE MOTOR PER WING. Each drives its own sector wheel through its own rope drive. Port and starboard wings on the same pair are driven by separate motors on axis0 and axis1 of one ODrive board. Confirm: 5010-14 designation (NOT 5010-6 which has different poles).	~$25 ea
8	pcs	Motor output drum for rope drive	Pressed or bolted to motor shaft. Sized so that motor_rps × (drum_radius / sector_radius) = 7:1 ratio. Verify gear ratio: turn motor 7 times, sector turns once.	Custom

8	sets	Motor mount hardware	Bolts, standoffs, vibration isolators. Apply medium thread lock (Loctite 243) to all motor mount fasteners. Re-check torque after first 10 flights.	Hardware

3. ODRIVE MOTOR CONTROLLERS — 4 BOARDS × 2 AXES = 8 MOTORS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
4	pcs	ODrive v3.6 motor controller	Each board drives TWO independent motors: axis0 and axis1. Board 0 → axis0=W1 port, axis1=W1 stbd. Board 1 → axis0=W2 port, axis1=W2 stbd. Board 2 → axis0=W3 port, axis1=W3 stbd. Board 3 → axis0=W4 port, axis1=W4 stbd. CAN node IDs: 0,1,2,3. vel_limit=150 rad/s (set via mco_odrive_setup.py). Both axes configured: pole_pairs=7, MOTOR_TYPE_HIGH_CURRENT.	~$130 ea
1	pcs	MCP2515 or MCP2518FD SPI-to-CAN adapter	Connects Raspberry Pi SPI0 to the CAN bus. MCP2515: 8 MHz oscillator, interrupt on GPIO25, SPI0 CE0. MCP2518FD: supports CAN-FD (faster). Either works for MCO-1D. Wiring: Pi SPI0 → adapter → ODrive 0 → ODrive 1 → ODrive 2 → ODrive 3.	~$5
2	pcs	CAN bus 120 Ω termination resistors	One at each END of the CAN bus chain — at Pi side and at ODrive 3. Between CAN-H and CAN-L. Required. Bus will not work without both.	<$1
2+	m	CAN bus cable — twisted pair	CAN-H and CAN-L must be twisted together. Daisy-chain: Pi adapter → ODrive 0 → 1 → 2 → 3 → 120Ω end resistor.	Per length

4. PITCH SERVOS — ANGLE OF ATTACK — 8 WINGS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	TORQ TCM-1001 high-speed digital servo (or equivalent)	MUST have analog feedback potentiometer output pin — this is essential for the per-wing AoA sensing system. 333 Hz PWM capable. 7.4 V supply from dedicated servo LiPo. Resolution: 0.1°. Controls AoA per wing independently. GPIO PWM pins: W1p=18, W1s=23, W2p=24, W2s=25, W3p=12, W3s=16, W4p=20, W4s=21.	~$15 ea
8	wires	Pitch servo feedback signal wires	From each servo feedback/pot output pin to MCP3208 #1 ADC input. Channel map: W1p→CH0, W1s→CH1, W2p→CH2, W2s→CH3, W3p→CH4, W3s→CH5, W4p→CH6, W4s→CH7. Keep wires under 30 cm. Add 1 kΩ series resistor at ADC input end.	<$1 total
16	pcs	1 kΩ resistor, 0.25 W (8 for pitch + 8 for sweep)	Series protection on each ADC input line. Prevents ADC input damage if wire accidentally contacts 5 V servo rail. Solder in-line on the signal wire near the ADC.	<$1 total

5. SWEEP SERVOS — FORE-AFT POSITION — 8 WINGS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	DS3225 high-torque digital servo (or equivalent)	MUST have analog feedback potentiometer output pin. 25 kg·cm torque. Digital. 7.4 V supply. Standard 50 Hz PWM. GPIO PWM pins: W1p=4, W1s=17, W2p=27, W2s=22, W3p=5, W3s=6, W4p=13, W4s=19. In normal flight holds a fixed position. In H-16 (figure-eight) and H-18 (sweep-clap) moves dynamically within each stroke.	~$10 ea
8	wires	Sweep servo feedback signal wires	From each servo feedback pin to MCP3208 #2 ADC input. Same channel map as pitch: W1p→CH0, W1s→CH1 ... W4s→CH7. 1 kΩ series resistor at ADC input (shared budget with pitch above).	<$1 total

6. PER-WING SENSING — MCP3208 ADC CHIPS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost

2	pcs	MCP3208-CI/P 8-channel 12-bit SPI ADC	Same chip used twice — one for pitch feedback, one for sweep. ADC #1 (PITCH): SPI Bus 1, CE1. Chip select = GPIO7 (SPI1 CE1). ADC #2 (SWEEP): SPI Bus 1, CE0. Chip select = GPIO8 (SPI1 CE0). Both share the same SPI1 bus (CLK=GPIO11, MOSI=GPIO10, MISO=GPIO9). VREF = 3.3 V from Pi. VDD = 3.3 V. GND to Pi GND. Resolution: 12-bit over 3.3 V → 0.8 mV/step → ~0.044°/step. All 8 channels read in ~107 µs — negligible at 50 Hz. /boot/config.txt addition: dtoverlay=spi1-2cs (enables SPI1 with CE0 and CE1 both active).	~$3 ea

7. FLIGHT COMPUTER — RASPBERRY PI 4

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	Raspberry Pi 4 Model B — 4 GB RAM	4 GB is required. 2 GB is insufficient for extended discovery flights where the knowledge registry and aero state engine together exceed 500 MB. Runs Wave Library at 50 Hz. Mean tick: ~1.1 ms dev, ~13 ms on Pi 4. 1.5× headroom against 20 ms budget.	~$55
1	pcs	MicroSD card — 64 GB, Class 10 / A1 endurance rating	High-endurance card recommended (frequent writes at 50 Hz). Flight log grows at ~50 MB per hour. 64 GB = 1,280 flight hours. Never store logs on tmpfs — data lost on power cycle.	~$12
1	pcs	Pi heatsink + case with airflow	Motor vibration environment. Must be rigidly mounted. Active cooling (small fan) recommended for summer operations and extended Discovery Mode sessions.	~$10

8. SENSORS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	MPU-6050 IMU — I2C address 0x68	Primary attitude reference. 6-DOF accelerometer + gyroscope at 100 Hz. Provides roll, pitch, yaw rates and linear accelerations. Used by stall detection Tier 1 (vibration correlation) and Tier 4 (lift waveform). Connect to Pi I2C1 (SDA=GPIO2, SCL=GPIO3).	~$3
1	pcs	BMP280 barometer — I2C address 0x76	Altitude above the launch point. Used when h_agl > 3 m where the lidar rangefinder loses terrain accuracy. Same I2C1 bus as IMU.	~$3
1	pcs	TF-Luna lidar rangefinder — 50 Hz	Actual height above ground or water surface, 0.1–8 m range. Primary height reference for all landing and approach operations. Over water: reads actual wave surface height for wave estimation. Mount facing straight down, clear of wing flapping arc. Connect UART or I2C per module variant.	~$20

1	pcs	DS18B20 waterproof temperature sensor	Outside air temperature (OAT) for density altitude computation. Required for high-altitude SAR operations. One-Wire bus on GPIO4. 4.7 kΩ pull-up resistor to 3.3 V required. /boot/config.txt addition: dtoverlay=w1-gpio,gpiopin=4 Verify after reboot: cat /sys/bus/w1/devices/28-*/w1_slave	~$3
1	pcs	HX711 ADC + strain gauge load cell (50 N minimum)	Static thrust measurement during TC-01 bench test and TC-02 hover. NOT mounted in flight — bench fixture only. HX711 DT pin → GPIO26. HX711 SCK pin → GPIO7. Requires mco_loadcell.py (to be written before TC-01).	~$12

9. RADIO AND NAVIGATION

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	RC transmitter + receiver, SBUS or IBUS, minimum 4 channels	Channel assignment: CH1=Roll, CH2=Pitch, CH3=Lever, CH4=Yaw. VERIFY mapping before TC-02. Wrong mapping = loss of control. SBUS output from receiver → Pi UART (GPIO14 TX, GPIO15 RX). Bind receiver to transmitter before first power-on.	Per system
1	pcs	Pixhawk flight controller (any model)	Used as SENSOR BRIDGE only. Does NOT run flight control. Provides GPS position, compass heading, and additional IMU data to the Pi via MAVLink protocol. Connect Pixhawk TELEM2 → Pi UART at 921,600 baud. mco_mavlink.py reads this in a background thread.	~$40
1	pcs	GPS module for Pixhawk	External GPS + compass. Mount as far from motors as possible. Used by Discovery Mode for racetrack pattern positioning.	~$20
1	pcs	Touchscreen display — 3.5 inch minimum	Runs mco_panel.py via mco-panel.service. Shows 4 function buttons (PRE-FLIGHT, ARM, DISCOVERY, POST-FLIGHT) and 6 margin bars (altitude, stall, frequency, amplitude, gust, combined). Bars turn amber at 50% margin, red at 25%. Connect via SPI or HDMI.	~$25

10. POWER SYSTEM

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	Main LiPo — 4S, minimum 4,000 mAh, 30 C discharge	Powers all 8 motors through 4 ODrive boards. Larger capacity (5,000–6,000 mAh) recommended for Discovery Mode flights. Bus voltage is monitored from ODrive telemetry. LOW warning at 30% remaining. CRITICAL at 20%.	Per spec
1	pcs	Servo LiPo — 2S, 1,500 mAh, 7.4 V nominal	DEDICATED supply for all 16 servos (8 pitch + 8 sweep). Keeps motor PWM switching noise off the servo rail. Without dedicated supply: noise can cause servo position errors.	Per spec
1	pcs	BEC 5 V / 3 A regulated	Clean regulated Pi supply. Motor PWM switching WILL corrupt the SD card if Pi power is shared with the motor supply. Use a filtered, isolated BEC.	~$8

1	pcs	30 A blade fuse + holder — main battery lead	Protects wiring on hard short. Install as close to battery as possible.	~$3
1	pcs	5 A blade fuse + holder — Pi BEC output		~$2

11. COMMISSIONING TEST EQUIPMENT (TC-01 / TC-02)

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	Static thrust test fixture	Rigid frame to restrain aircraft for TC-01 thrust measurement. Must withstand full thrust (>40 N) without any movement. HX711 + load cell mounted in compression: aircraft pushes down on cell. Record thrust vs frequency from 1.5 Hz to 3.0 Hz in 0.25 Hz steps.	Custom
1	pcs	Infrared thermometer or thermal camera	Motor temperature monitoring during TC-01-11 frequency ceiling test. Target: below 60°C at operating frequency after 60 seconds. Check all 8 motors individually.	~$20
1	pcs	Laptop (any OS) with Python 3.8+	Runs mco_ground_station.py. WiFi or ethernet to Pi. Second operator position during all TC flights. Displays live telemetry, all margin bars, and per-wing sensing data.	Per availa bility
1	set	Field safety kit	LiPo-rated fire extinguisher. First aid kit. Safety glasses. Hearing protection for sustained bench runs. Minimum 30 m clear zone around aircraft during TC-02.	Per site

12. CONSUMABLES AND TOOLS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	roll	Thread lock — medium strength (Loctite 243 blue)	Apply to: all motor mount fasteners, all sector wheel axle fasteners, all ODrive mounting hardware. Do NOT apply to rope drive adjustment hardware (need to re-tension).	Per roll
1	can	Conformal coating (if water operations planned)	Apply to Pi board, all ODrive boards, HX711, servo electronics. Apply in thin coats, avoid connectors. Required before any water operations testing.	Per can
1	tube	Cyanoacrylate adhesive (thin)	Wing panel repairs, rope drive end sealing.	Per tube
2	pcs	Spare rope drive sections	Pre-cut and pre-tensioned. Rope is the most likely wear item. Fraying shows as rising motor current before failure.	Per length

13. SOFTWARE — ALL FREE / NO COST

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	file	mco_wave_library.py — 45,229 lines	Main flight control. 50 Hz tick stack (58 wrappers). 29 active hypotheses, 109 variants. Full optimizer, sensing, fault tolerance, aero state engine, discovery programme.	No cost
1	file	mco_main.py — 389 lines	Flight loop entry point. Reads sensors, calls tick(), reads ODrive positions and currents, reads MCP3208 ADCs, applies wing commands, logs data, sends telemetry.	No cost
1	file	mco_odrive_interface.py — 253 lines	4-board 8-axis ODrive CAN interface. Handles axis0 and axis1 independently per board. Reads position and current per wing.	No cost
1	file	mco_servo_interface.py — 110 lines	Servo PWM via pigpio.	No cost
1	file	mco_mavlink.py — 290 lines	Pixhawk MAVLink sensor bridge.	No cost
1	file	mco_gs_server.py — 193 lines	Ground station TCP server port 5760.	No cost
1	file	mco_ground_station.py — 660 lines	Laptop ground station. Shows sensing display panel with per-wing AoA, effective AoA, force waveform, trim balance, Strouhal, stall detection, servo health, and rope load.	No cost

1	file	mco_logger.py — 220 lines	50 Hz CSV logger. Captures all per-wing sensing fields: 8× AoA, 8× effective AoA, 8× stroke position, 8× current, 4× Strouhal, force waveform, trim balance, stall per pair, 8× servo health RMS.	No cost
1	file	mco_odrive_setup.py — 189 lines	ODrive configuration. Configures BOTH axis0 and axis1 on all 4 boards: pole_pairs=7, vel_limit=150, MOTOR_TYPE_HIGH_CURRENT.	No cost
1	file	mco_preflight.py — 782 lines	Three-tier pre-flight check system. Includes ADC feedback check and ODrive axis1 verification.	No cost
1	file	mco_panel.py — 485 lines	Touchscreen panel driver.	No cost
1	file	mco_hypothesis_tool.py — 347 lines	Hypothesis authoring. Contains registry for H-28 to H-32.	No cost
8	files	Other supporting files	mco_sync.py, mco_optimiser.py, mco_adaptive_optimiser.py, mco_intent_control.py, mco_taxi_park.py, mco_cg_estimator.py, mco_flight_test_protocol.py, mco_first_run_wizard.py.	No cost
1	file	mco_loadcell.py	PENDING — HX711 reader for TC-01 static thrust. Write before TC-01.	No cost
1	file	Raspberry Pi OS Lite 64-bit	From raspberrypi.com/software — download Raspberry Pi Imager. Use Lite (no desktop) for maximum headroom on the 20 ms tick budget.	No cost

SUMMARY — COMPLETE HARDWARE COUNT

Total motors	8 (one per wing, all independent)
Total ODrive boards	4 (2 axes each = 8 motor axes total)
Total pitch servos	8 (one per wing, feedback pin required)
Total sweep servos	8 (one per wing, feedback pin required)
Total servos	16
Pitch ADC	1 × MCP3208 on SPI1 CE1 (8 channels)
Sweep ADC	1 × MCP3208 on SPI1 CE0 (8 channels)
1 kΩ resistors	16 (8 pitch + 8 sweep feedback lines)
CAN bus	4 ODrives in chain, 120 Ω at each end
Flight computer	Raspberry Pi 4, 4 GB
Sensors	IMU + baro + lidar + OAT + HX711 load cell
Stroke sensing	ODrive pos_estimate — no extra hardware
Radio	RC receiver + Pixhawk sensor bridge
Wave Library	45,229 lines
Hypotheses	29 active, 109 variants
GS commands	114
Discovery modes	10
Tick time (Pi 4 est)	~13 ms mean — 1.5× headroom
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■ WARNING: Each wing has its own independent motor and sector wheel. There is NO shared rope drive between port and starboard on any pair. 8 wings = 8 independent mechanical assemblies.

✓ Stroke position sensing (per-wing flap angle) is provided by reading ODrive axis0.pos_estimate and axis1.pos_estimate on all 4 boards via the existing CAN bus. No new hardware is required for stroke sensing.

✓ Both MCP3208 ADC chips share SPI Bus 1. CE1 = pitch servos, CE0 = sweep. Enable both with one line in /boot/config.txt: dtoverlay=spi1-2cs